GÓMEZ'S COMPOUND GRAPHS ΩH_q AND THE DIAMETER-8 ENTRIES OF THE BIPARTITE DEGREE–DIAMETER TABLE

DEGREE–DIAMETER LAB

Abstract. Six diameter-8 entries of the bipartite degree–diameter table, credited to Delorme, Gómez and Quisquater, are the orders of the compound graphs ΩH_q of Gómez's 1986 thesis ($q = 7, 8, 9, 11, 11, 13$), which the thesis states to have diameter ≤ 8 without proof. We show that, as defined (every copy of the hexagon carries the same labels), ΩH_q has diameter at least 9 under every reading of the figures (Theorems 1 and 2; Theorem 1 is also formalised in Lean). Giving each copy its own labelling is the natural repair; we show that it fails for every $q \geq 7$ whenever the labelling respects the geometry of the hexagon (Theorems 3 and 4). Arbitrary relabellings remain open for $q \geq 4$; at $q = 2, 3$ they do give diameter 8.

Labels. Every claim carries one of these labels.

- **Theorem**: proved by hand in this note, with a complete proof. Lemmas and corollaries proved here carry the tag "Theorem" in their heading. *Theorem (computer-assisted)*: the proof is complete except for one named Computer-checked statement.
- **Computer-checked**: a statement about finite data, with the programs and logs that check it (§8). "Twice" means by two programs written separately, in different sessions, sharing no code.
- **Heuristic**: an argument with a named gap. It is not a proof.
- **Fact**: a standard result, quoted with a reference.

File paths refer to the repository; the branch of each source is listed in §8.

Summary

In the 2013 survey table [4, Table 4], reproduced on the CombinatoricsWiki [1], six entries with $D = 8$ are the orders of Gómez's graphs ΩH_q [3, §2.9.3, values on p. 46]. Here $H(q)$ is the split Cayley hexagon, its incidence graph has $N_1 = 2N$ vertices with $N = (q^6 - 1)/(q - 1)$, and $q + 3 = 2\alpha + \beta$ with $\beta \in \{0, 1\}$:

cell	table value	graph	q	α	copies	hubs	order $5(\alpha+1)N_1$
$(10, 8)$	1,176,480	ΩH_7	7	5	25	196,080	1,176,480
$(11, 8)$	2,246,940	ΩH_8	8	5	25	374,490	2,246,940
$(12, 8)$	4,650,100	ΩH_9	9	6	30	664,300	4,650,100
$(14, 8)$	14,172,480	ΩH_{11}	11	7	35	1,771,560	14,172,480
$(15, 8)$	14,172,480	ΩH_{11} (reused)	11	7	35	1,771,560	14,172,480
$(16, 8)$	36,201,060	ΩH_{13}	13	8	40	4,022,340	36,201,060

The thesis states $D \leq D_1 + 2 = 8$ and gives no proof. The paper the table cites [2] was never published, as far as we could find.

(1) **Theorem 1 (one label; also in Lean).** If a new vertex is joined only to copies of one vertex t of B_1 and misses a copy, it is at distance $\geq d_{B_1}(t, v) + 3$ from the copy of v in the missed copy. In one reading of the figures this gives $\operatorname{diam} \Omega H_q \geq 9$ for every q and every pairing of labels.

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(2) **Theorem 2 (two labels, $q \geq 7$).** For the thesis's labelling and every reading of the figures, every hub has at least $q^5 - 5q^4 - 8q^3 - 12q^2 - 16q - 10$ vertices at distance ≥ 9 in every copy it misses. This covers the six table entries. By computer, ΩH_7 and ΩH_9 have diameter exactly 9 for the patterns and pairings of C2(a).

(3) **Theorems 3 and 4 (geometric relabellings, $q \geq 7$).** Give every copy its own labelling, and call it geometric if every transfer map between two copies met by one hub class is induced by an automorphism of the incidence graph of $H(q)$. For every $q \geq 7$, every geometric labelling, every reading and every mix of block types, the diameter is still ≥ 9. For $q \geq 10$ the proof is by hand; for $q = 8, 9$ it uses a computer count of route words (C4); for $q = 7$ it also uses Bonferroni inequalities and a character-sum bound. (For q not a power of 3 there are no geometric labellings at all in reading 1 with both block types, so there the statement is empty and Theorem 2 is what applies.)

(4) **Open (§6).** Arbitrary (non-geometric) relabellings for every $q \geq 4$ (at $q = 2, 3$ they give diameter 8); labellings chosen per hub class and copy; other wirings; the geometric class at $q \leq 5$.

(5) **What can be claimed (§7).** The six entries have no checkable graph of diameter 8. We do not claim that no $(\Delta, 8)$ graph of these orders exists.

1. The split Cayley hexagon

1.1. **Coordinates.** We use the model of our diameter-7 note [5, §1.1], which all of the lab's generators use. Let q be a prime power, $\mathbb{F} = \mathbb{F}_q$ and $V = \mathbb{F}^7$ with coordinates $(X_0, \ldots, X_6)$. Put

$$\mathsf{Q}(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2,$$
$$b(X, Y) = \mathsf{Q}(X + Y) - \mathsf{Q}(X) - \mathsf{Q}(Y) = X_0Y_4 + X_4Y_0 + X_1Y_5 + X_5Y_1 + X_2Y_6 + X_6Y_2 - 2X_3Y_3.$$

The parabolic quadric $Q(6, q)$ is the set of points $\langle x \rangle$ of $\mathrm{PG}(6, q)$ with $\mathsf{Q}(x) = 0$. Two points $\langle x \rangle, \langle y \rangle$ of $Q(6, q)$ are *perpendicular*, $x \perp y$, if $b(x, y) = 0$; since $\mathsf{Q}(\lambda x + \mu y) = \lambda\mu\, b(x, y)$, this means that the line xy lies on $Q(6, q)$. We write $x^\perp$ for the hyperplane $b(x, \cdot) = 0$.

The split Cayley hexagon $H = H(q)$ [6], [7, §2.4] has

- points: the set P of all $N = (q^6 - 1)/(q - 1)$ points of $Q(6, q)$;
- lines: the set L of the lines of $Q(6, q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ (for two spanning points x, y) satisfy

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31};$$

- incidence: containment.

For example, the $q + 1$ lines through $e_0 = (1, 0, \ldots, 0)$ are $\langle e_0, \lambda e_5 + \mu e_6 \rangle$, $(\lambda : \mu) \in \mathrm{PG}(1, q)$; they lie in the totally singular plane $\langle e_0, e_5, e_6 \rangle$. The incidence graph $I = I(q)$ has vertex set $P \cup L$ ($2N$ vertices). We write d for its distance, $S_i(z) = \{u : d(z, u) = i\}$, $B(z, R) = \{u : d(z, u) \leq R\}$, and $\mathrm{opp}(z) = S_6(z)$ (the elements *opposite* z).

Fact 1.1 ([6], [7, §2.4])**.** *$H(q)$ is a generalized hexagon of order (q, q): I is connected, bipartite, $(q+1)$-regular, of diameter 6 and girth 12. The lines of $H(q)$ through a point x are the lines through x of a totally singular plane $\pi_x \subset x^\perp$.*

Fact 1.2 ([6], [7])**.** *The automorphisms of I fixing P are the collineations of $H(q)$; they form $\mathrm{G}_2(q) \rtimes \mathrm{Aut}(\mathbb{F}_q)$ and are induced by semilinear maps of V preserving Q. For q prime every collineation is induced by a linear map in $\mathrm{G}_2(q) \leq \mathrm{SO}(7, q)$. For example, for $A \in \mathrm{SL}(3, q)$ the map $(v, a, w) \mapsto (Av, a, A^{-T}w)$, where $x = (v, a, w)$ with $v = (x_0, x_1, x_2)$, $a = x_3$, $w = (x_4, x_5, x_6)$, lies in $\mathrm{G}_2(q)$. An automorphism of I exchanging P and L (a* duality*) exists if and only if q is a power of 3.*

Fact 1.3 (eigenvalues in G_2)**.** *For $h \in \mathrm{G}_2(q)$ acting on V, the eigenvalues of h in $\overline{\mathbb{F}}_q$, with multiplicity, are $1, x_1^{\pm1}, x_2^{\pm1}, x_3^{\pm1}$ with $x_1x_2x_3 = 1$. (The semisimple part of h is conjugate over $\overline{\mathbb{F}}_q$ into the maximal torus $\mathrm{diag}(a,b,c,1,a^{-1},b^{-1},c^{-1})$, $abc = 1$, of the subgroup $\mathrm{SL}(3)$ above, and h has the characteristic polynomial of its semisimple part.)*

1.2. **Distances.**

Lemma 1.4 (Theorem; distance profile)**.** *Let t, u be vertices of I with $d(t,u) = k$.*

- *(a) If $1 \le k \le 5$, exactly one neighbour of u is at distance $k-1$ from t, and the other q are at distance $k+1$. If $k = 6$, all $q+1$ neighbours of u are at distance 5 from t.*
- *(b) A walk $(u_0, u_1, \dots, u_i)$ with $u_{j+1} \neq u_{j-1}$ for all j (*non-backtracking*) and $i \le 6$ has $d(u_0,u_i) = i$. For $i \le 5$ each vertex of $S_i(u_0)$ is the end of exactly one such walk. Hence $|S_i(u)| = s_i := (q+1)q^{i-1}$ for $1 \le i \le 5$, $|S_6(u)| = q^5$; we put $s_0 := 1$.*

Proof. (a) I is bipartite, so the neighbours of u are at distance $k \pm 1$ from t; since the diameter is 6, they are all at distance 5 when $k = 6$. For $1 \le k \le 5$ at least one is at distance $k-1$ (the previous vertex of a geodesic). If two were, there would be two distinct geodesics from t to u, and their union would contain a cycle of length $\le 2k \le 10 < 12$. (b) By induction on j, using (a) with $t := u_0$: if $(u_0, \dots, u_j)$ is a geodesic and $1 \le j \le 5$, then u_{j-1} is the unique neighbour of u_j at distance $j-1$ from u_0, so $u_{j+1} \neq u_{j-1}$ is at distance $j+1$. For $i \le 5$ the geodesic from u_0 to a vertex of $S_i(u_0)$ is unique by the same girth argument. The count follows, and $|S_6(u)| = 2N - 1 - \sum_{i=1}^{5}(q+1)q^{i-1} = q^5$. □

Lemma 1.5 (Theorem; orthogonality [7, §2.4])**.** *For points x, y of $H(q)$: $d(x,y) \le 4$ if and only if $x \perp y$. So $\mathrm{opp}(x) \cap P = P \setminus x^{\perp}$.*

Proof. If $d(x,y) \le 4$, there is a point z with x and y on lines of H through z ($z = x$ if $d(x,y) \le 2$). These lines lie in the totally singular plane π_z (Fact 1.1), so $b(x,y) = 0$. Conversely, $x^{\perp} \cap Q(6,q)$ is a cone with vertex x over a $Q(4,q)$, with $1 + q(q^3+q^2+q+1)$ points, and by Lemma 1.4 there are $1 + s_2 + s_4 = 1 + q(q+1) + q^3(q+1)$ points at distance ≤ 4 from x. The numbers are equal, so the inclusion is an equality. □

Example. $\mathrm{opp}(e_0) \cap P = \{\langle x \rangle \in Q(6,q) : x_4 \neq 0\} = \{\langle (x_3^2 - x_1x_5 - x_2x_6,\ x_1,\ x_2,\ x_3,\ 1,\ x_5,\ x_6) \rangle : x_1, x_2, x_3, x_5, x_6 \in \mathbb{F}\}$, q^5 points, since $b(e_0, x) = x_4$.

Lemma 1.6 (Theorem; opposite elements in balls)**.** *Let t, z be vertices of I of the same type and $d = d(t,z)$. Then*

$$|B(z,4) \cap \mathrm{opp}(t)| = \begin{cases} 0 & z = t, \\ q^4 & d \in \{2,4,6\}, \end{cases} \qquad |B(z,2) \cap \mathrm{opp}(t)| = \begin{cases} 0 & d \in \{0,2\}, \\ q^2 & d \in \{4,6\}. \end{cases}$$

Proof. The elements of $B(z,4)$ of the type of t are at distance $i \in \{0,2,4\}$ from z. By Lemma 1.4(b) the elements of $S_i(z)$ correspond to the non-backtracking walks $z = h_0, h_1, \dots, h_i$, and we follow $d(t, h_j)$ along the walk with Lemma 1.4(a): from h_j at distance $k \in [1,5]$ from t, if h_{j-1} is the neighbour at distance $k-1$ ("we came up"), all q choices of h_{j+1} go up to $k+1$; if h_{j-1} is at distance $k+1$ ("we came down"), one choice goes down to $k-1$ and $q-1$ go up; from distance 6 all choices go down to 5.

- $d = 0$: $B(t,4)$ misses $\mathrm{opp}(t)$.
- $d = 2$: four steps reach distance 6 only if every step goes up: q ways for the first step (avoiding the one down-neighbour of z) and q^3 for the others; so q^4 elements of $S_4(z)$, and none of $B(z,2)$, are opposite t.
- $d = 4$: $i = 2$: both steps up, q^2. $i = 4$: if h_1 goes down (one way, to distance 3), h_2 must go up ($q-1$ ways; going down reaches distance 2) and then h_3, h_4 go up (q^2 ways): $(q-1)q^2$. If

h_1 goes up (q ways, to 5), h_2 goes up to 6 (q ways), h_3 comes down to 5 (q ways), and $h_4 \neq h_2$ must go up; h_3 has one neighbour at distance 4 and h_2 is at distance 6, so $q-1$ ways: $q^3(q-1)$. In total $q^2 + (q-1)q^2 + (q-1)q^3 = q^4$, and $B(z,2)$ gives q^2.

- $d = 6$: $i = 0$ gives 1. $i = 2$: h_1 is at distance 5 ($q+1$ ways) and $h_2 \neq z$; one neighbour of h_1 is at distance 4 and z is at 6, so $q-1$ ways to reach 6: $(q+1)(q-1)$. $i = 4$: if h_2 is the neighbour of h_1 at distance 4, then $h_3 \neq h_1$ goes up to 5 in $q-1$ ways, and $h_4 \neq h_2$ goes up to 6 in all q ways (h_2 is the only neighbour of h_3 at distance 4): $(q+1)(q-1)q$. If h_2 is at distance 6 ($q-1$ ways), h_3 is at 5 (q ways) and $h_4 \neq h_2$ reaches 6 in $q-1$ ways: $(q+1)(q-1)q(q-1)$. So $i = 4$ gives $(q+1)(q-1)q^2$, and $1 + (q^2-1) + (q^4-q^2) = q^4$; $B(z,2)$ gives $1 + q^2 - 1 = q^2$. □

Remark. The same numbers, for every radius ≤ 4 and both types, follow from the intersection array $\{q+1, q, q, q, q; 1, 1, 1, 1, 1, q+1\}$ of the distance-regular graph I; the referee of the relabelling note computed them that way (`research/referee/omega-repair/logs/lemma31_drg.txt` on its branch). For this note we also counted them directly on $H(3)$ and $H(4)$ built by `hexq6.c`, together with Lemma 1.5 on $H(3)$ (`lem16.py`).

2. Gómez's construction

2.1. **The text.** J. Gómez Martí, *Diámetro y vulnerabilidad en redes de interconexión*, PhD thesis, UPC, 1986 [3]. We read the scan on F. Comellas's site (161 pages, 8,989,821 bytes; from thesis page 33 on, PDF page = thesis page + 6). The construction is §2.9.3, pp. 43–45; verbatim:

> p. 43: "2.9.3 En esta construcción, que da lugar a grafos bipartitos, también se toman, como en los Apartados 2.8.4 y 2.8.9, copias de K_1, (vértices aislados). Sin embargo la estructura de interconexión es algo distinta de las anteriores en el sentido de que el diámetro resultante no está acotado por $D_0 + D_1 + 1 = D_1 + 1$ sino por $D_1 + 2$."
>
> p. 44: "Tomemos primeramente 5 copias de un grafo bipartito B_1 con diámetro D_1 par y 10 conjuntos de N_1 vértices cada uno. Cada 10 vértices —uno de cada conjunto— se unen a las copias de B_1 según muestra la Figura 2.21.a, [figure] de manera que su grado queda igual a dos. Ahora incrementamos dicho grado hasta 4, uniendo estos vértices a 5 copias de B_1 según la Figura 2.21.b.
>
> El grafo bipartito así construido tiene $N = 10\frac{N_1}{2} + 10N_1 = 15N_1$ vértices, grado $\Delta = \max\{4, \Delta_1 + 2\}$ y diámetro $D \leq D_1 + 2$. En general, si $\Delta_1 + 2 = 2\alpha + \beta$, $\alpha \geq 2$, $\beta = 0, 1$, se toman 5α copias de B_1 de manera, que, utilizando cualquiera de las dos estructuras anteriores el grado de los vértices correspondientes a las copias de K_1 se incrementa hasta 2α. Entonces se obtiene un grafo bipartito, que denotamos por ΩB_1, con parámetros $N = 10\frac{N_1}{2} + 5\alpha N_1 = 5(\alpha+1)N_1$, $\Delta = \Delta_1 + 2$ y $D \leq D_1 + 2$."

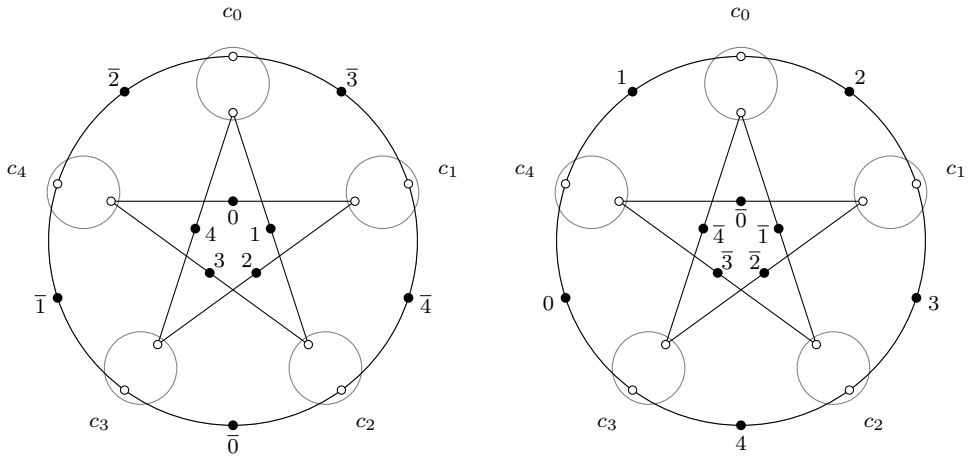
In English: take 5 copies of a bipartite graph B_1 of even diameter D_1 and 10 sets of N_1 vertices each (the count shows that $N_1/2$ is meant); every 10 vertices, one from each set, are joined to the copies as in Fig. 2.21a, so that their degree is 2; then the degree is raised to 4 by joining them to 5 more copies as in Fig. 2.21b. In general, if $\Delta_1 + 2 = 2\alpha + \beta$, take 5α copies, using either structure, so that the copies of K_1 get degree 2α; the result ΩB_1 has $N = 5(\alpha+1)N_1$, $\Delta = \Delta_1 + 2$ and $D \leq D_1 + 2$.

No proof is given. §2.8.2 (p. 30) says the proofs of this part are omitted: "prescindiendo de las demostraciones que, aunque son largas, no ofrecen ninguna dificultad pues se basan en las mismas ideas desarrolladas hasta ahora". The list on p. 46 includes ΩH_5: $N(8,8) \geq 195\,300$; ΩH_7: $N(10,8) \geq 1\,176\,480$; ΩH_8: $N(11,8) \geq 2\,246\,940$; ΩH_9: $N(12,8) \geq 4\,650\,100$; ΩH_{11}: $N(14,8) \geq 14\,172\,480$; $\Omega H_{11}d$: $N(15,8) \geq 14\,978\,080$; ΩH_{13}: $N(16,8) \geq 36\,201\,060$. Here H_q is the incidence graph of $H(q)$: $D_1 = 6$, $\Delta_1 = q+1$, $N_1 = 2N$. $\Omega H_{11}d$ is not defined in §§2.7–2.9; the table's $(15,8)$ value is $|\Omega H_{11}|$.

2.2. **Labels.** §2.7.1 (p. 21): "Supongamos que los vértices del conjunto V_{ij} están rotulados con los elementos del conjunto $\mathbb{Z}/n_i\mathbb{Z}$." Both colour classes of B_1 are labelled by $\mathbb{Z}/n_1\mathbb{Z}$, $n_1 = N_1/2$, and

every adjacency rule of §§2.7–2.9 is written in these labels, the same in every copy. §2.8.9 (p. 39), which §2.9.3 cites for its copies of K_1, defines them as "(t,i), $t \in \{0,1,\dots,n_1-1\}$, $i \in \{0,1,\dots,4\}$", and Fig. 2.18 (p. 41) joins vertex $(t,1,j,k,h)$ of copy (B_1,k,h) to (t,i), as in "$(t,1,0,0,0) \sim (t,0)$": **a copy of K_1 with label t is joined only to vertices labelled t**, in either colour class. For $B_1 = I(H(q))$ we write the labelling as two bijections $P_0\colon \mathbb{Z}/N\mathbb{Z} \to P$ (the point labelled t) and $L_0\colon \mathbb{Z}/N\mathbb{Z} \to L$ (the line labelled t). The thesis does not fix them; $\sigma = L_0 P_0^{-1}$ (point $t \mapsto$ line t) is an arbitrary bijection $P \to L$.

2.3. **The figures.** Both figures put five copies of B_1 at the corners of a pentagon, inside a circle and a pentagram. Each copy is drawn with a dashed line (its two colour classes) and two dots: an *outer* dot on the circle and an *inner* dot at a corner of the pentagram. Every K_1-vertex of the block is a dot on an arc or on a pentagram edge, joined to the two dots at its ends. In Fig. 2.21a the K_1-vertex k lies on the pentagram edge between copies $k-1$ and $k+1$, and $\overline{k}$ on the arc between copies $k+2$ and $k+3$; in Fig. 2.21b, k lies on the arc between copies $k+3$ and $k+4$, and $\overline{k}$ on the pentagram edge between copies $k-1$ and $k+1$ (indices mod 5). Figure 1 redraws them.



type a (Fig. 2.21a) type b (Fig. 2.21b)

FIGURE 1. One block of each type (positions $c_0,\dots,c_4$ clockwise from the top). Open circles: the outer mark (on the circle) and the inner mark (pentagram corner) of one row t in each copy. Black dots: the hubs (t,k), labelled k, and $(t,\overline{k})$, labelled $\bar{k}$. Each hub is adjacent to the two marks at the ends of its edge or arc.

The figures do not say which colour class the inner dot is. In Figs. 2.3, 2.6 and 2.18 the class above the dashed line is class $j=0$, and in both Figs. 2.21 the outer dot of the top copy is above the dashed line. With class 0 = points this gives **reading 1**: outer = points and inner = lines in every copy. **Reading 2** (used in the first computations): unbarred K_1-vertices always on points and barred ones always on lines. The thesis does not say which blocks are of type (a) and which of type (b) ("utilizando cualquiera de las dos estructuras"), so we allow any pattern. Below we allow the inner class to be chosen freely in every copy; this contains both readings and their duals.

2.4. **Definition with exact neighbourhoods.**

Definition 2.1 (ΩH_q)**.** The data are a prime power q and $\alpha = \lfloor (q+3)/2 \rfloor$; a block pattern $(\tau_0,\dots,\tau_{\alpha-1}) \in \{a,b\}^\alpha$; an inner class $\iota(c) \in \{P,L\}$ for every copy c; and a *labelling*: bijections $P_c\colon \mathbb{Z}/N\mathbb{Z} \to P$ and $L_c\colon \mathbb{Z}/N\mathbb{Z} \to L$ for every copy c.

- *Copies.* $c = 5B + i$ with block $B \in \{0,\dots,\alpha-1\}$ and position $i \in \mathbb{Z}_5$; 5α copies. Copy vertices (c,x), $x \in P \cup L$, with $(c,x) \sim (c,y)$ when x and y are incident.

- *Hubs* (the copies of K_1). (t,k) and $(t,\overline{k})$ for $t \in \mathbb{Z}/N\mathbb{Z}$ (the *row*) and $k \in \mathbb{Z}_5$. The hub *classes* are $k = \{(t,k)\}_t$ (unbarred) and $\overline{k} = \{(t,\overline{k})\}_t$ (barred).
- *Copies met by a class* (positions mod 5):

$$C(k) = \{5B+k-1,\ 5B+k+1 : \tau_B = a\} \cup \{5B+k+3,\ 5B+k+4 : \tau_B = b\},$$
$$C(\overline{k}) = \{5B+k+2,\ 5B+k+3 : \tau_B = a\} \cup \{5B+k-1,\ 5B+k+1 : \tau_B = b\}.$$

- *Where a class sits.* In a copy c of a block of type a, unbarred classes sit on the inner class $\iota(c)$ and barred classes on the outer class; in a block of type b, unbarred classes sit on the outer class and barred classes on the inner class.
- *Hub edges.* For $c \in C(Y)$: $(t,Y) \sim (c, P_c(t))$ if Y sits on points in c, and $(t,Y) \sim (c, L_c(t))$ if Y sits on lines in c.

The **thesis labelling** is $P_c = P_0$, $L_c = L_0$ for all c. **Reading 1**: $\iota(c) = L$ for all c. **Reading 2**: $\iota(c) = P$ in a-blocks and L in b-blocks (then unbarred hubs sit on points everywhere). Gómez's ΩH_q is the thesis labelling with any pattern and either reading.

Exact neighbourhoods. Write $m_c(t)$ for the inner mark of row t in copy c ($P_c(t)$ if $\iota(c) = P$, $L_c(t)$ if $\iota(c) = L$) and $\bar{m}_c(t)$ for the outer mark. For $c = 5B + i$:

vertex	neighbours
(c,x)	(c,y) for the $q+1$ elements y incident with x, and two hubs:
$\tau_B = a$, $x = m_c(t)$	$(t, i-1)$, $(t, i+1)$
$\tau_B = a$, $x = \bar{m}_c(t)$	$(t, \overline{i+2})$, $(t, \overline{i+3})$
$\tau_B = b$, $x = m_c(t)$	$(t, \overline{i-1})$, $(t, \overline{i+1})$
$\tau_B = b$, $x = \bar{m}_c(t)$	$(t, i+1)$, $(t, i+2)$
(t,k)	$(c, m_c(t))$ for $c = 5B+k\pm1$ in blocks with $\tau_B = a$;
	$(c, \bar{m}_c(t))$ for $c = 5B+k+3,\ 5B+k+4$ in blocks with $\tau_B = b$
$(t,\overline{k})$	$(c, \bar{m}_c(t))$ for $c = 5B+k+2,\ 5B+k+3$ in blocks with $\tau_B = a$;
	$(c, m_c(t))$ for $c = 5B+k\pm1$ in blocks with $\tau_B = b$

Lemma 2.2 (Theorem; parameters)**.** *For every pattern, inner classes and labelling, ΩH_q is a simple bipartite graph on $10(\alpha+1)N = 5(\alpha+1)N_1$ vertices, with $5\alpha N(q+1) + 20\alpha N$ edges; copy vertices have degree $q+3$ and hubs degree 2α, so $\Delta = q+3 = \Delta_1 + 2$. Every hub meets 2α of the 5α copies and misses 3α.*

Proof. The copy at position i of an a-block is met by the unbarred classes $i \pm 1$ and the barred classes $i+2, i+3$; of a b-block, by the unbarred classes $i+1, i+2$ and the barred classes $i \pm 1$. So each copy is met by exactly two classes of each bar, the two classes of one bar sit on the same type, and each class has one hub per row. Every element of a copy is the mark of exactly one row of its type, so it gets exactly two hubs. A class meets two distinct copies in each block. For the colouring, give unbarred hubs colour 0; in every copy, the type on which the unbarred classes sit colour 1 and the other type colour 0; barred hubs colour 1. Copy edges join a point and a line, so every edge has ends of different colours. □

3. Theorem 1: one label (also in Lean)

Theorem 1 (projection)**.** *Let B be a graph and G a graph whose vertices are of two kinds:* copy vertices *v, each with a copy index $c(v) \in J$ and a label $f(v) \in V(B)$, and* new vertices *x, each with a label $f(x) \in V(B)$. Assume*

(h1) if $u \sim w$ are copy vertices, then $c(u) = c(w)$ and $f(u) \sim f(w)$ in B;

(h2) every neighbour w of a new vertex x is a copy vertex with $f(w) = f(x)$;

(h3) no two new vertices are adjacent.

If x is a new vertex with no neighbour in copy j, and v is a copy vertex with $c(v) = j$, then $d_G(x, v) \geq d_B(f(x), f(v)) + 3$.

Proof. Let $(x = w_0, w_1, \ldots, w_\ell = v)$ be a walk. By (h2), w_1 is a copy vertex with $f(w_1) = f(x)$. Let m be the number of new vertices among $w_1, \ldots, w_{\ell-1}$. A new vertex w_i $(0 < i < \ell)$ has copy vertices w_{i-1}, w_{i+1} on both sides (h2, h3), with $f(w_{i-1}) = f(w_i) = f(w_{i+1})$; every other edge $w_i w_{i+1}$ $(i \geq 1)$ joins two copy vertices and maps to an edge of B by (h1). Deleting the m new vertices and one of the two equal labels around each, the labels of $w_1, \ldots, w_\ell$ form a walk in B from $f(x)$ to $f(v)$ of length $\ell - 1 - 2m$. So $\ell \geq d_B(f(x), f(v)) + 1 + 2m$. If $m = 0$, then $w_1, \ldots, w_\ell$ are copy vertices joined by copy edges, so by (h1) $c(w_1) = c(v) = j$ and w_1 is a neighbour of x in copy j, a contradiction. So $m \geq 1$. □

Lean. `BipartiteDD.LeanMaxx.Projection.edist_ge` in `lean/BipartiteDD/LeanMaxx/Projection/Main.lean` (walk lemmas in `Walk.lean`). Copies are encoded by `c : V → Option ι` (`none` = new vertex) and labels by `f : V → β`:

```
theorem edist_ge
    (h1 : ∀ u w i j, G.Adj u w → c u = some i → c w = some j → i = j ∧ B.Adj (f u) (f w))
    (h2 : ∀ x w, G.Adj x w → c x = none → c w ≠ none ∧ f w = f x)
    (h3 : ∀ x w, G.Adj x w → c x = none → c w = none → False)
    {x v : V} (hx : c x = none) {j : ι} (hmiss : ∀ w, G.Adj x w → c w ≠ some j)
    (hv : c v = some j) :
    (B.dist (f x) (f v) + 3 : ℕ∞) ≤ G.edist x v
```

The corollary `ediam_ge` gives `(D₁ + 3 : ℕ∞) ≤ G.ediam` when `B.edist t s = D₁`, `f x = t`, `f v = s`. `#print axioms` gives only `propext`, `Classical.choice`, `Quot.sound`. An independent session rebuilt the modules (Lean v4.34.1, Mathlib as pinned), regenerated the axioms output byte for byte, searched the files for `sorry`, `admit`, `native_decide` and similar, and read the statement against the specification (`research/lean-maxx/VERIFY.md`). The Lean statement is Theorem 1 as stated here (`B.dist` is 0 when `f x` and `f v` are not connected, which only weakens it; `VERIFY.md` calls it more general than the first write-up's version, which assumed B bipartite of even diameter). That ΩH_q satisfies (h1)–(h3) is checked in Corollary 3.1; it is not part of the Lean files.

Corollary 3.1 (Theorem)**.** *Take the thesis labelling and inner classes for which every hub class sits on one type in all copies it meets (reading 2; its dual; reading 1 when all blocks have one type). Then for every q, every pattern and every pairing σ: for every hub x and every copy j that x misses, the q^5 vertices (j, y) with y opposite the mark of x are at distance ≥ 9 from x. So* $\operatorname{diam} \Omega H_q \geq 9$. *More generally, for any bipartite graph B_1 with two vertices at distance D_1, ΩB_1 in reading 2 has diameter $\geq D_1 + 3$.*

Proof. Put $c((c, y)) = c$ and $f((c, y)) = y$, and label a hub by its mark ($P_0(t)$ or $L_0(t)$, the same in all copies by assumption). Copy edges satisfy (h1), hub edges (h2) by Definition 2.1, and there are no hub–hub edges. Every hub misses $3\alpha \geq 1$ copies (Lemma 2.2), and every vertex of I has q^5 opposite vertices (Lemma 1.4). Apply Theorem 1. For general B_1: in reading 2 the unbarred classes sit on all labels of class 0 and the barred classes on all labels of class 1, so whichever class holds the pair, some hub carries one of its labels. □

For instance, if $P_0(t) = e_0$, the hub (t, k) (reading 2) is at distance ≥ 9 from every point $\langle (x_3^2 - x_1x_5 - x_2x_6, x_1, x_2, x_3, 1, x_5, x_6) \rangle$ of each of the 3α copies it misses (Example in §1).

Computer-checked C1 (the hexagons)**.** *The incidence graphs of $H(q)$ used in the computations were checked by all-pairs BFS (no symmetry assumed): $2N$ vertices, bipartite, $(q+1)$-regular, diameter 6, girth 12. Three times:* `hexq6.c` *(this model) with* `apsp_mt`*, $q = 3, 4, 5, 7, 8, 9, 11, 13$ (diameter 6 checked; girth 12 then follows from the bipartite Moore bound);* `hexgen.c`*,* `hexgf.c`

(split octonions) with `bitapsp.c`, $q = 2, 3, 4, 5, 7, 8, 9, 11, 13$, *and* `buildhex.py` *(split octonions) with* `ghcheck.c`, $q = 3, 7, 8, 9$ *(both with sphere sizes from every vertex). A* $(q+1)$*-regular bipartite graph on* $2N$ *vertices with diameter* 6 *and girth* 12 *is the incidence graph of a generalized hexagon of order* (q, q).

Computer-checked C2 (the thesis graphs)**.** *(a)* Exact diameters *(both block types present).* ΩH_3 *and* ΩH_5 *(*195,300 *vertices, the thesis's* $(8, 8)$ *value):* 9 *by all-pairs BFS in both readings,* σ *the identity and two random* σ*, patterns aba, aab, abab, aabb (referee of the first write-up); independently, by BFS from orbit representatives under checked automorphisms,* ΩH_3 *in both readings and* ΩH_5 *in reading 2 (abab) and reading 1 (abab with a Singer-equivariant and a random* σ*; aabb) (first write-up).* ΩH_7*:* 9 *in reading 2 (ababa) and reading 1 (ababa, random* σ*), by BFS from* 117,648 *sources, one in every orbit of the rotation and the block swaps, each automorphism checked edge by edge.* ΩH_9*:* 9 *in reading 2 (aaabbb;* 18 *orbits of a group generated by the rotation, block swaps, a Singer cycle and a transvection of* $H(9)$*, all checked on the* 27,900,600 *edges) and in reading 1 (Singer-equivariant* σ*,* 4,380 *orbits); again* 9 *in reading 2 and in reading 1 with a random* σ *from* 398,580 *orbit representatives (second program).*

(b) All six thesis values. *BFS from a hub gives eccentricity* 9 *for* ΩH_5*,* ΩH_7*,* ΩH_8*,* ΩH_9*,* ΩH_{11}*,* ΩH_{13} *at the thesis orders, in both readings (reading 2 by the first write-up and its referee; reading 1 by the referee and the table audit). In reading 2 the vertices at distance* 9 *from a hub* (t, k) *are exactly the points opposite* t *in the* 3α *copies it misses:* $3\alpha q^5 = 252{,}105$ *(*$q = 7$*),* 491,520 *(*8*),* 1,062,882 *(*9*),* 3,382,071 *(*11*),* 8,911,032 *(*13*).*

(c) One block type*: diameter* 10 *(all pairs at* $q = 2, 3, 5$*; at* $q = 9$ *by BFS from orbit representatives under checked automorphisms; at* $q = 7$ *a direct BFS gives eccentricity* 10*).*

(d) Explicit pairs at distance 9 *for every cell:* `checks/witness_omega_thesis.txt` *(table audit),* `research/d12D8/omega/witness_pairs.txt` *(own-12-8).*

4. Theorem 2: two labels

In reading 1 with both block types, an unbarred hub (t, k) sits on line $L_0(t)$ in a-blocks and on point $P_0(t)$ in b-blocks, so Theorem 1 does not apply. The following count covers every reading.

Theorem 2. *Let* $q \geq 7$*. Take* ΩH_q *with the thesis labelling, any block pattern and any inner classes (so both readings and every pairing* σ*). Let* x *be a hub and* j *a copy that* x *misses. Then at least*

$$q^5 - (A + B + 1) = q^5 - 5q^4 - 8q^3 - 12q^2 - 16q - 10 > 0$$

vertices of copy j *are at distance* ≥ 9 *from* x*, where (sums over integers* $e_i \geq 0$*)*

$$A = \sum_{e_1+e_2\in\{0,2,4\}} s_{e_1}s_{e_2} = 5q^4 + 8q^3 + 6q^2 + 4q + 2, \qquad B = \sum_{e_1+e_2+e_3\leq 2} s_{e_1}s_{e_2}s_{e_3} = 6q^2 + 12q + 7.$$

In particular $\operatorname{diam} \Omega H_q \geq 9$.

q	A	B	$A + B + 1$	q^5	far vertices per missed copy ($\geq$)
7	15,073	385	15,459	16,807	1,348
8	24,994	487	25,482	32,768	7,286
9	39,161	601	39,763	59,049	19,286
11	84,625	865	85,491	161,051	75,560
13	161,449	1,177	162,627	371,293	208,666

Proof. Let τ be the involution of $P \cup L$ with $\tau(P_0(s)) = L_0(s)$ for every row s. Every hub of row s is adjacent, in each copy it meets, to $P_0(s)$ or to $L_0(s)$. Let x have row t. By Lemma 2.2, in copy j one type $T \in \{P, L\}$ has the colour of the neighbours of x; let $z = P_0(t)$ if $T = P$ and

$z = L_0(t)$ if $T = L$. We show that at most $A + B + 1$ elements $w \in \mathrm{opp}(z)$ have $d(x, (j, w)) \le 8$; as $|\mathrm{opp}(z)| = q^5$, the others are at distance ≥ 9.

Let $w \in \mathrm{opp}(z)$; it has type T. The graph is bipartite and (j, w) has the colour of the neighbours of x, so $d(x, (j, w))$ is odd; suppose it is ≤ 7 and take a shortest path. It leaves x to a vertex (c_1, a_0) with $a_0 \in \{z, \tau(z)\}$, then alternates walks inside copies with passes copy $\to$ hub $\to$ copy. A pass through a hub of row s goes from (c, a) to (c', a') with $a, a' \in \{P_0(s), L_0(s)\}$, so $a' = a$ (a *stay*) or $a' = \tau(a)$ (a *switch*). Since x misses j there is at least one pass. With e copy edges and $p \ge 1$ passes the length is $1 + e + 2p \le 7$ and odd, so e is even, $e \le 4$ and $p \le 3 - e/2$. Let $r \le p$ be the number of switches. Merging the stays, the labels form a walk in I from a_0 with e edges in total, with τ applied r times in between, ending at w. Every edge of I and every switch changes the type, and w has the type of z, so r is even if $a_0 = z$ and odd if $a_0 = \tau(z)$. The set of ends of walks of length f from a vertex u is $\bigcup S_g(u)$ over $g \le f$, $g \equiv f \pmod 2$, of size at most the sum of the s_g. So:

- $a_0 = z$, $r = 0$: w ends a walk of length $e \le 4$ from z, so $d(z, w) \le 4$ and $w \notin \mathrm{opp}(z)$.
- $a_0 = z$, $r = 2$: $p \ge 2$, so $e \le 2$; the label walk is $z \to (f_1$ edges$) \to u_1$, switch, $\to (f_2) \to u_2$, switch, $\to (f_3) \to w$, with $f_1 + f_2 + f_3 = e \le 2$. At most B such w.
- $a_0 = \tau(z)$, $r = 1$: $\tau(z) \to (f_1) \to u_1$, switch, $\to (f_2) \to w$ with $f_1 + f_2 = e \in \{0, 2, 4\}$. At most A such w.
- $a_0 = \tau(z)$, $r = 3$: $p = 3$, $e = 0$, and $w = \tau^4(z) = z \notin \mathrm{opp}(z)$ (counted as 1 to match the earlier reports).

The closed forms are expansions (for A: $s_0 s_0 + 2 s_0 s_2 + s_1 s_1 + 2 s_0 s_4 + 2 s_1 s_3 + s_2 s_2$). Finally, for $q \ge 7$, $q^4(q - 5) \ge 2q^4 \ge 14q^3$, so $q^5 - 5q^4 - 8q^3 - 12q^2 - 16q - 10 \ge 6q^3 - 12q^2 - 16q - 10 \ge 30q^2 - 16q - 10 > 0$. □

Remark. The proof uses only: the graph is bipartite; all copies carry the labels P_0, L_0; every hub of row s is adjacent, in each copy it meets, only to $P_0(s)$ or $L_0(s)$; there are no hub–hub edges; x misses a copy. It does not use that x meets both types, so it also covers the hubs to which Theorem 1 applies. (This applies the two corrections asked for by the referee of the first write-up: "bipartite" belongs in the hypothesis, and the appeal to Theorem 1 for single-label hubs is not needed.) For $q \le 6$ the count is not enough ($A + B + 1 > q^5$); see C3.

Corollary 4.1 (Theorem)**.** *As defined in the thesis (every copy carries the same labels, and a copy of K_1 with label t is joined only to vertices labelled t), ΩH_q has diameter ≥ 9 for $q = 7, 8, 9, 11, 13$, under every reading of Figs. 2.21a/b, every pattern of block types and every pairing of point labels with line labels. These are the six table entries.*

Computer-checked C3 (small q; slack in Theorem 2)**.** *The count of Theorem 2 fails for $q \le 6$. There, all-pairs BFS gives diameter 9 with both block types for ΩH_2 (pattern ab, three σ; table audit), ΩH_3 and ΩH_5 (both readings, several σ; C2(a)); $q = 4$ was not computed with the thesis labelling. Under reading 1 (σ the identity or random), the largest number of opposite vertices of the right type within distance 7 of a hub, in a copy it misses, was 7,562–8,039 at $q = 7$ (bound 15,459), 13,950 at $q = 8$ (25,482), 22,771–22,825 at $q = 9$ (39,763), 52,372 at $q = 11$ (85,491) and 104,225 at $q = 13$ (162,627).*

5. Relabelling the copies

Diameter ≥ 9 comes from the shared labels. A labelling that differs from copy to copy is not in the thesis (§§2.7–2.9 use one labelling for all copies), but it is the obvious repair, and at $q = 2, 3$ it works (C7). This section shows that it cannot work at the table's q with labellings that respect the geometry.

5.1. **The model.** A labelling (P_c, L_c) as in Definition 2.1 gives each copy one point labelling and one line labelling, used by every hub class of the corresponding type that meets the copy. For a hub class Y and copies $c, d \in C(Y)$, the **transfer** $T_Y(c \to d)$ maps the mark of row t in c to the mark of row t in d: it is $P_d P_c^{-1}$, $L_d L_c^{-1}$, $L_d P_c^{-1}$ or $P_d L_c^{-1}$, according to the types Y sits on in c and d.

Definition 5.1. A labelling is **geometric** if every transfer is the restriction (to P or to L) of an automorphism of I: a collineation of $H(q)$ or, when q is a power of 3, a duality.

Labellings chosen per pair (class, copy), in which the two classes of one bar meeting a copy may use different bijections, are outside this model even when every transfer is a collineation; see §6.

Lemma 5.2 (Theorem; one automorphism per copy)**.** *Relabelling every copy c of a graph ΩH_q by an automorphism g_c of I (replacing every vertex (c, y) by $(c, g_c(y))$) gives an isomorphic graph. Applied to the thesis graph: for collineations g_c this is the labelling $P_c = g_c P_0$, $L_c = g_c L_0$; for dualities, the labelling $P_c = g_c L_0$, $L_c = g_c P_0$ with the inner class of c exchanged. Such a graph is isomorphic to the thesis graph it came from, so its diameter is ≥ 9 by Corollary 3.1 (if that graph is in reading 2: every q) and by Corollary 4.1 (every reading, $q \geq 7$).*

Proof. Copy edges are preserved because g_c is an automorphism of I; hub edges are carried along by definition. □

Lemma 5.3 (Theorem; reduction and normal form)**.** *Take a geometric labelling, with any pattern and inner classes.*

- *(a) If q is not a power of 3, every hub class sits on one type in all copies it meets, all unbarred classes sit on the same type and all barred classes on the other: the inner classes are those of reading 2 or of its dual.*
- *(b) If q is a power of 3, the graph is isomorphic to one with a geometric labelling in reading 2 (same pattern).*
- *(c) In reading 2 and in its dual, the graph is isomorphic to one with $P_c = P_0$ for all c and $L_c = \lambda_c L_0$ with collineations λ_c (the* normal form*). We call the bar that sits on points the* ***point bar*** *$\mathcal{P}$ (unbarred in reading 2, barred in its dual) and the other the* ***line bar*** *$\mathcal{L}$.*

Proof. (a) A transfer from a copy where Y sits on points to one where it sits on lines maps P to L; an automorphism of I doing that is a duality, and there is none (Fact 1.2). In every copy the two unbarred classes sit on the same type. The copies are linked by unbarred classes: inside a block, the pairs $\{k \pm 1\}$ (type a) or $\{k+3, k+4\}$ (type b), $k \in \mathbb{Z}_5$, form a 5-cycle on the positions, and every class meets every block. So all unbarred classes sit on one type, and the barred classes sit on the other type in every copy. (b) Fix a duality δ and relabel by δ every copy in which the unbarred classes sit on lines (Lemma 5.2). The new transfers are the old ones composed with $\delta^{\pm 1}$, so they are still automorphisms of I, and now all unbarred classes sit on points. (c) The transfers of the point bar are restrictions of collineations to P, and that bar links all copies (as in (a); for the barred classes the pairs are $\{k+2, k+3\}$ and $\{k \pm 1\}$). Fixing a copy c_0, every $P_c P_{c_0}^{-1}$ is a product of transfers, so $P_c = a_c P_{c_0}$ with a collineation a_c; likewise $L_c = b_c L_{c_0}$. Relabel copy c by a_c^{-1} (Lemma 5.2): P_c becomes P_{c_0} and L_c becomes $a_c^{-1} b_c L_{c_0}$. Unbarred rows and barred rows are different vertices, so renaming the rows of each bar separately gives $P_c = P_0$ and $L_c = \lambda_c L_0$ with $\lambda_c = a_c^{-1} b_c$. □

From now on we use the normal form, identify the rows of the point bar with points through P_0 (so the hub (t, X), $X \in \mathcal{P}$, is adjacent to the point t of every copy in $C(X)$), and let the collineations λ_c act on P and on L.

5.2. **Routes reach balls.** For $X \in \mathcal{P}$ and $c' \notin C(X)$ let $N(X, c')$ be the set of copies $c_1 \in C(X)$ such that some class of $\mathcal{L}$ meets both c_1 and c'.

Lemma 5.4 (Theorem; routes)**.** *Let $X \in \mathcal{P}$, $c' \notin C(X)$ and $t \in P$. Every point v of copy c' with $d((t,X),(c',v)) \le 7$ lies in*

$$B(t,4) \;\cup \bigcup_{c_1 \in N(X,c')} B(\lambda_{c'}\lambda_{c_1}^{-1}(t),4) \;\cup \bigcup_{(c,d)\in W_2} B(\lambda_c\lambda_d^{-1}(t),2),$$

where W_2 is the set of ordered pairs of distinct copies of three kinds: (c',c_2) with c_2 sharing a class of $\mathcal{P}$ with some $c_1 \in C(X)$ and a class of $\mathcal{L}$ with c'; (c',c_1) with $c_1 \in C(X)$ and some c_2 sharing a class of $\mathcal{L}$ with c_1 and one with c'; (c_2,c_1) with $c_1 \in C(X)$ and c_2 sharing a class of $\mathcal{L}$ with c_1 and a class of $\mathcal{P}$ with c'.

Proof. A path from $h = (t,X)$ to (c',v) goes to the point t of a copy $c_1 \in C(X)$, then alternates walks inside copies, of lengths $a_1,\dots,a_r \ge 0$, with $r-1$ passes copy $\to$ hub $\to$ copy, ending in c'. A hub meets each copy at most once, so consecutive copies differ; $c_1 \ne c'$, so $r \ge 2$. The length is $1 + \sum a_i + 2(r-1) \le 7$, so $\sum a_i \le 8 - 2r$ and $r \le 4$. A pass through a $\mathcal{P}$-hub happens at a point and keeps it ($\mathcal{P}$-hubs sit on the same point in every copy). A pass through an $\mathcal{L}$-hub (m,Y) happens at a line and goes from $\lambda_c L_0(m)$ in c to $\lambda_d L_0(m)$ in d: it applies $\lambda_d\lambda_c^{-1}$. Collineations preserve distances, so by induction the end lies in $B(g(t),\sum a_i)$, where g is the product of the maps applied. The walk starts and ends at points and an $\mathcal{L}$-pass happens at a line, so the segments next to $\mathcal{L}$-passes are odd and $\sum a_i$ is even.

- $r = 2$ ($\sum a_i \le 4$): a $\mathcal{P}$-pass gives $g = \mathrm{id}$; an $\mathcal{L}$-pass needs a class of $\mathcal{L}$ meeting c_1 and c', that is $c_1 \in N(X,c')$, and gives $g = \lambda_{c'}\lambda_{c_1}^{-1}$.
- $r = 3$ ($\sum a_i \le 2$): $\mathcal{P}$ then $\mathcal{P}$ gives id. $\mathcal{P}$ then $\mathcal{L}$ forces $a_1 = 0$, $a_2 = a_3 = 1$ and $g = \lambda_{c'}\lambda_{c_2}^{-1}$ (first kind). $\mathcal{L}$ then $\mathcal{P}$ forces $a_1 = a_2 = 1$, $a_3 = 0$ and $g = \lambda_{c_2}\lambda_{c_1}^{-1}$ (third kind). $\mathcal{L}$ then $\mathcal{L}$ forces $a_1 = a_3 = 1$, $a_2 = 0$: both passes use the same row m, so $g = \lambda_{c'}\lambda_{c_2}^{-1}\lambda_{c_2}\lambda_{c_1}^{-1} = \lambda_{c'}\lambda_{c_1}^{-1}$ (second kind).
- $r = 4$: all $a_i = 0$, so the walk never reaches a line, all passes are $\mathcal{P}$-passes, and $v = t$. □

The line bar. Relabelling every copy c by λ_c^{-1} (Lemma 5.2) gives $L_c = L_0$ for all c and $P_c = \lambda_c^{-1}P_0$. So the same statement holds with points and lines exchanged and λ_c replaced by λ_c^{-1}: for $X \in \mathcal{L}$, $c' \notin C(X)$ and a row m, every line of c' within distance 7 of (m,X) lies in the corresponding union of balls around $\ell = L_0(m)$, the lines $\lambda_{c'}^{-1}\lambda_{c_1}(\ell)$ (radius 4) and $\lambda_c^{-1}\lambda_d(\ell)$ (radius 2), where $N(X,c')$ and W_2 are defined as above with the roles of $\mathcal{P}$ and $\mathcal{L}$ exchanged. (The relabelling note states this in its Lemma 2.1.)

For a pair (X,c'), X in either bar, let $r_4 = |N(X,c')|$, and let m_2 be the number of pairs in W_2 that are not of the form (c',c_1) with $c_1 \in N(X,c')$. Different pairs may give the same map; counting pairs only overestimates.

Corollary 5.5 (Theorem)**.** *Every hub of X has at least $q^5 - r_4q^4 - m_2q^2$ vertices of c' at distance ≥ 9: points if $X \in \mathcal{P}$, lines if $X \in \mathcal{L}$. If $r_4 = \alpha$, then $m_2 \le (4\alpha - 1) + 2\alpha \cdot 2(2\alpha - 1) = 8\alpha^2 - 1$.*

Proof. Take $X \in \mathcal{P}$ (for $\mathcal{L}$ exchange points and lines; Lemma 1.6 holds for both types). By Lemma 1.6, $B(t,4)$ misses $\mathrm{opp}(t)$, each other radius-4 ball meets it in at most q^4 points and each radius-2 ball in at most q^2. The remaining points of $\mathrm{opp}(t)$ in c' are at distance > 7 from (t,X), and at odd distance, since all points of all copies have one colour; so at distance ≥ 9. For the bound on m_2: the first two kinds are pairs (c',b) with $b \ne c'$, at most $5\alpha - 1 - r_4 = 4\alpha - 1$ of them outside the radius-4 words; the third kind has $\le 2\alpha$ choices of c_1 and, since c_1 is met by two classes of $\mathcal{L}$ that meet 2α copies each, $\le 2(2\alpha - 1)$ choices of c_2. □

5.3. **Starved pairs.** Call (X,c'), with X in either bar and $c' \notin C(X)$, **starved** if $r_4 = \alpha$. In the pairs of Lemma 5.6, exactly one copy of $C(X)$ per block shares a class of the other bar with c' (that every pair has $r_4 \ge \alpha$ is C4, and is not used in the proofs).

Lemma 5.6 (Theorem)**.** *For every block pattern there are starved pairs with X unbarred and starved pairs with X barred:*

- *$X = k$ unbarred: $c' = 5B' + k + 3$ for a block B' of type a; if all blocks are of type b, $c' = 5B' + k$.*
- *$X = \overline{k}$ barred: $c' = 5B' + k + 3$ for a block B' of type b; if all blocks are of type a, $c' = 5B' + k + 1$.*

Proof. The classes meeting the copy at position i are: in an a-block, unbarred $i \pm 1$ and barred $i+2, i+3$; in a b-block, unbarred $i+1, i+2$ and barred $i \pm 1$. For $X = k$ ($\mathcal{L}$ = barred): c' at position $k+3$ of an a-block has barred classes $\{k, k+1\}$. In an a-block, $C(k)$ has positions $k-1$ (barred $\{k+1, k+2\}$: shares) and $k+1$ (barred $\{k+3, k+4\}$: does not); in a b-block, positions $k+3$ (barred $\{k+2, k+4\}$: no) and $k+4$ (barred $\{k, k+3\}$: shares). If all blocks are of type b, c' at position k has barred $\{k-1, k+1\}$, and $C(k)$ has $k+3$ (barred $\{k+2, k+4\}$: shares $k+4 = k-1$) and $k+4$ (barred $\{k, k+3\}$: no). For $X = \overline{k}$ ($\mathcal{L}$ = unbarred): c' at position $k+3$ of a b-block has unbarred $\{k+4, k\}$. In an a-block, $C(\overline{k})$ has $k+2$ (unbarred $\{k+1, k+3\}$: no) and $k+3$ (unbarred $\{k+2, k+4\}$: shares); in a b-block, $k-1$ (unbarred $\{k, k+1\}$: shares) and $k+1$ (unbarred $\{k+2, k+3\}$: no). If all blocks are of type a, c' at position $k+1$ has unbarred $\{k, k+2\}$, and $C(\overline{k})$ has $k+2$ (unbarred $\{k+1, k+3\}$: no) and $k+3$ (unbarred $\{k+2, k+4\}$: shares). In each case $c' \notin C(X)$. □

Computer-checked C4 (route words)**.** *For $\alpha = 5, 6, 7, 8$ and every block pattern (by block swaps only the number of a-blocks matters), the routes of Lemma 5.4 were enumerated for every pair (X, c'):*

- *every pair has $r_4 \geq \alpha$, and no route ends in a radius-0 ball other than the identity;*
- *for $\alpha = 5$ ($q = 7, 8$), every starved pair, of either bar and in every pattern, has $m_2 \leq 140$;*
- *for $\alpha = 6$ ($q = 9$), every starved pair has $m_2 \leq 201$;*
- *in every pattern some starved pair (of one of the two bars) has $m_2 \leq 134$ ($\alpha = 5$; equality for aabbb, aaabb), ≤ 197 ($\alpha = 6$; aaabbb), ≤ 262 ($\alpha = 7$; aaabbbb, aaaabbb), ≤ 347 ($\alpha = 8$; aaaabbbb).*

The last item was found by `routecount.c` *and* `routecount2.c` *of the relabelling note and by* `myroutes.py`, `sweep.py` *of its referee, which enumerates walks symbolically. For this note all four items were re-run for $\alpha = 5, 6, 7, 8$ by two programs:* `starved.py` *(with the referee's* `myroutes.py`*) and* `c4_families.py` *(written by the independent reviewer of this note from the families of Lemma 5.4 and Definition 2.1). They agree; the largest m_2 of a starved pair is* 140, 201, 272, 353.

5.4. $q \geq 8$.

Theorem 3 (computer-assisted for $q = 8, 9$)**.** *Let $q \geq 8$. For every block pattern, every choice of inner classes (both readings, their duals, or any choice per copy) and every geometric labelling, ΩH_q has diameter ≥ 9. More precisely, in the normal form, for every starved pair (X, c') every hub (t, X) has at least $q^5 - \alpha q^4 - m_2 q^2 > 0$ vertices of c' at distance ≥ 9.*

Proof. By Lemma 5.3 we may assume the normal form, in reading 2 or its dual. Take a starved pair (Lemma 5.6) and apply Corollary 5.5. For $q \geq 10$, $m_2 \leq 8\alpha^2 - 1$ and the bound is positive when $q^2(q-\alpha) \geq 8\alpha^2$; with $\alpha \leq (q+3)/2$ it suffices that $q^2(q-3) \geq 4(q+3)^2$, that is $q^3 - 7q^2 - 24q - 36 \geq 0$, which holds at $q = 10$ (value 24) and beyond (the derivative $3q^2 - 14q - 24$ is positive). For $q = 8$ ($\alpha = 5$), C4 gives $m_2 \leq 140$ for every starved pair, and the bound is $\geq 32{,}768 - 20{,}480 - 8{,}960 = 3{,}328$; for $q = 9$ ($\alpha = 6$), $m_2 \leq 201$ and the bound is $\geq 59{,}049 - 39{,}366 - 16{,}281 = 3{,}402$. □

q	α	far vertices per hub of a starved class	best pair of either bar, worst pattern (C4)
8	5	$\geq$ 3,328 for every starved pair (uses C4)	3,712
9	6	$\geq$ 3,402 for every starved pair (uses C4)	3,726
11	7	$\geq$ 11,253 (by hand, $m_2 \leq 391$)	26,862
13	8	$\geq$ 56,446 (by hand, $m_2 \leq 511$)	84,162

This covers $(11,8)$, $(12,8)$, $(14,8)$, $(15,8)$, $(16,8)$. Different starved pairs give disjoint sets of (hub, vertex) pairs; summing over the starved pairs of both bars, every geometric labelling leaves at least $3.33 \cdot 10^9$ far pairs at $q = 8$ and $7.43 \cdot 10^9$ at $q = 9$ (Computer-checked three times: the relabelling note, its referee, and `c4_families.py`); with the point bar alone, at least $6.2 \cdot 10^8$ and $1.2 \cdot 10^9$. The relabelling note first treated the other readings separately for $q = 8, 9, 11, 13, 27$; its referee observed that Lemma 5.3(b) reduces them to reading 2 for every q, which is how they are handled here.

5.5. $q = 7$**.** At $q = 7$ the bound of Corollary 5.5 is negative ($16{,}807 - 5 \cdot 2{,}401 - 140 \cdot 49 = -2{,}058$), so the overlaps of the five radius-4 balls must be used. Here q is odd and b is nondegenerate; 7 is prime and not a power of 3, so the collineations are elements of $\mathrm{G}_2(7) \leq \mathrm{SO}(7,7)$ (Fact 1.2) and the reading is reading 2 or its dual (Lemma 5.3(a)). Fix a starved pair (X, c') of the point bar and write $g_1, \ldots, g_5$ for the elements $\lambda_{c'}\lambda_{c_i}^{-1}$, $c_i \in N(X, c')$, and $y_i = g_i(t)$. By Lemma 1.5, $B(y,4) \cap P = y^\perp \cap Q(6,q)$ for a point y, so with $C_i := y_i^\perp \cap \mathrm{opp}(t)$ the points of c' opposite t within distance 7 of (t, X) lie in $C_1 \cup \cdots \cup C_5$ and in at most $m_2 \leq 140$ balls of radius 2.

Lemma 5.7 (Theorem; overlaps, q odd)**.** *Let y, y', t be distinct points of $Q(6,q)$.*

- *(a) If $y \perp y'$ and t lies on the line yy', then $y^\perp \cap y'^\perp \cap \mathrm{opp}(t) = \emptyset$. Otherwise t is not on the line yy' and $|y^\perp \cap y'^\perp \cap \mathrm{opp}(t)| \geq q^3 - q$.*
- *(b) If y, y', y'' are points not on one line, $|y^\perp \cap y'^\perp \cap y''^\perp \cap Q(6,q)| \leq (q+1)^2$.*
- *(c) If y, y', y'' lie on one totally singular line ℓ not through t, all pairwise and triple intersections of $y^\perp \cap \mathrm{opp}(t)$, $y'^\perp \cap \mathrm{opp}(t)$, $y''^\perp \cap \mathrm{opp}(t)$ equal $E_\ell := \ell^\perp \cap \mathrm{opp}(t)$, and $|E_\ell| \geq q^3 - q$.*

Proof. If $y \perp y'$ the line yy' is totally singular, and if t is on it, $y^\perp \cap y'^\perp = (yy')^\perp \subseteq t^\perp$. If $y \not\perp y'$, the line yy' meets $Q(6,q)$ only in y, y', so t is not on it. In the remaining cases $y^\perp \cap y'^\perp$ is the perp of a line, of vector dimension 5: a parabolic $Q(4,q)$ (if $y \not\perp y'$) or a cone with vertex the line yy' over a conic (if $y \perp y'$); both have $q^3 + q^2 + q + 1$ points. Those not opposite t are the points of $\Pi^\perp$, $\Pi = \langle y, y', t\rangle$ a plane. For three independent singular vectors the Gram matrix of b on Π has zero diagonal, so its rank is 3, 2 or 0 (a symmetric matrix with zero diagonal and rank 1 would be vv^T with all $v_i^2 = 0$). The radical of $\Pi^\perp$ is $\Pi \cap \Pi^\perp$, of dimension 0, 1 or 3. In the first case $\Pi^\perp$ (dimension 4) carries a nondegenerate form with $(q+1)^2$ or q^2+1 singular points; in the second it is a cone over a conic, $q^2 + q + 1$ points; in the third Π is totally singular and $\Pi^\perp = \Pi \oplus \langle w\rangle$ with $\mathsf{Q}(w) \neq 0$ ($Q(6,q)$ has no totally singular subspace of vector dimension 4), so its singular points are those of Π, $q^2 + q + 1$. So $|\Pi^\perp \cap Q(6,q)| \leq (q+1)^2$, which gives (a) and, with $\Pi = \langle y, y', y''\rangle$, (b). (c): $y^\perp \cap y'^\perp = \ell^\perp \subseteq y''^\perp$, and $|E_\ell| \geq q^3 - q$ by (a). □

Lemma 5.8 (Theorem; coincidences; uses $m_2 \leq 140$ from C4)**.** *If two of $t, y_1, \ldots, y_5$ coincide, the hub (t, X) has at least $q^5 - 4q^4 - 140q^2 = 16{,}807 - 9{,}604 - 6{,}860 = 343$ vertices of c' at distance ≥ 9.*

Proof. At most four distinct centres other than t remain; argue as in Corollary 5.5. □

Now let $t, y_1, \ldots, y_5$ be distinct. Call y_i, y_j **aligned** if they lie on one totally singular line through t (then $C_i \cap C_j = \emptyset$ by Lemma 5.7(a)); the aligned classes partition $\{y_1, \ldots, y_5\}$. A **rich line** is a maximal set of at least three y_i on one line of $\mathrm{PG}(6,q)$ not through t. Three collinear singular

points span a totally singular line, so every collinear triple of y's lies in an aligned class or in a rich line; a rich line meets each aligned class at most once, and two rich lines share at most one point.

Lemma 5.9 (Theorem; the configuration types at $q = 7$)**.** *Let A_2 be the number of aligned pairs, T_3 the number of triples meeting three different aligned classes, and $R_3 = \sum \binom{m}{3}$ over the rich lines (m points each). Then*

$$|C_1 \cup \cdots \cup C_5| \le U := 5q^4 - (10 - A_2 - R_3)(q^3 - q) + (T_3 - R_3)(q+1)^2,$$

except when one rich line holds all five points, where $U := 5q^4 - 4(q^3 - q)$. Classified by the sizes of the aligned classes and of the rich lines there are exactly 17 kinds of configuration (18 up to relabelling: the row $2,1,1,1 \mid 3$ holds two, with the rich line through the three singletons or through one point of the pair; they have the same A_2, T_3, R_3 and p). For $q = 7$ they are listed in Table 1, with $F = 16{,}807 - U - 6{,}860$ and the number p of pairs among $\{t, y_1, \ldots, y_5\}$ that are perpendicular in every configuration of the type, $p = \sum_{s \ge 2} \big(s + \binom{s}{2}\big)$ over aligned classes of size s plus $\sum \binom{m}{2}$ over rich lines. In the 9 rows with $F \le 0$, $p \ge 6$.

aligned classes	rich lines	A_2	T_3	R_3	U	F	p
5	–	10	0	0	12,005	−2,058	15
4, 1	–	6	0	0	10,661	−714	10
3, 2	–	4	0	0	9,989	−42	9
3, 1, 1	–	3	3	0	9,845	102	6
3, 1, 1	3	3	3	1	10,117	−170	9
2, 2, 1	–	2	4	0	9,573	374	6
2, 2, 1	3	2	4	1	9,845	102	9
2, 2, 1	3, 3	2	4	2	10,117	−170	12
2, 1, 1, 1	–	1	7	0	9,429	518	3
2, 1, 1, 1	3	1	7	1	9,701	246	6
2, 1, 1, 1	3, 3	1	7	2	9,973	−26	9
2, 1, 1, 1	4	1	7	4	10,517	−570	9
1, 1, 1, 1, 1	–	0	10	0	9,285	662	0
1, 1, 1, 1, 1	3	0	10	1	9,557	390	3
1, 1, 1, 1, 1	3, 3	0	10	2	9,829	118	6
1, 1, 1, 1, 1	4	0	10	4	10,373	−426	6
1, 1, 1, 1, 1	5	0	10	10	10,661	−714	10

TABLE 1. The 17 kinds of configuration of Lemma 5.9 at $q = 7$ ($q^4 = 2{,}401$, $q^3 - q = 336$, $(q+1)^2 = 64$, $140q^2 = 6{,}860$).

Proof. $|C_i| = q^4$ (Lemma 1.6, $y_i \ne t$). By Bonferroni, $|\bigcup C_i| \le S_1 - S_2 + S_3$, with S_k the sum of the k-fold intersections. By Lemma 5.7: an aligned pair contributes 0 to S_2, and every triple containing one contributes 0 to S_3; a pair or triple inside a rich line ℓ contributes $|E_\ell|$; every other pair contributes $\ge q^3 - q$, and every other triple (it is not collinear) $\le (q+1)^2$. A rich line with $m \in \{3, 4\}$ points contributes $-(\binom{m}{2} - \binom{m}{3})|E_\ell| = -2|E_\ell| \le -2(q^3 - q)$. Collecting terms gives U. If all five points lie on one rich line, all pairwise intersections equal E_ℓ, so $|\bigcup C_i| = 5q^4 - 4|E_\ell|$ exactly. The types: two 3-subsets of a 5-set that share at most one point share exactly one, and a third 3-subset would share two points with one of them; a 4- or 5-point rich line leaves no room for another. So the systems of rich lines are: none, one 3-line, two 3-lines through a common point, one 4-line, one 5-line. A rich line takes at most one point from each aligned class, which leaves

exactly the 17 rows (partitions 5, 41, 32: no rich line; 311: none or one 3-line; 221: none, one, or two 3-lines through the singleton; 2111: none, 3, $3+3$, 4; 11111: none, 3, $3+3$, 4, 5). The numbers A_2, T_3, R_3 and p depend only on the row. □

The table was also produced by three programs, which also classify by sizes: `degen.py` of the relabelling note, `mytypes.py` of its referee, and `types43.py`, written from scratch for this note; the reviewer of this note counted the 18 classes up to relabelling. The referee also built 2,500 explicit configurations in $Q(6,7)$ and found no union above the bound of its type.

Lemma 5.10 (Theorem; few points are moved to perpendicular points)**.** *Let q be odd with $3 \mid q-1$, and let $h \in \mathrm{G}_2(q)$ fix no point of $Q(6,q)$. Then* $\mathrm{Perp}(h) := \{s \in P : h(s) \perp s\}$ *has at most* $\frac{q^5-1}{q-1} + q^{3.5} + q^3 + (q-2)q^{1.5}$ *points: at most* 4,144 *of the* 19,608 *points when $q = 7$.*

Proof. Let $F(v) := b(v, hv)$ and $N_h := \#\{v \in V : \mathsf{Q}(v) = 0,\ F(v) = 0\}$; then $|\mathrm{Perp}(h)| = (N_h - 1)/(q-1)$. F is a quadratic form with polar form $b(v, (h+h^{-1})w)$, since $b(w, hv) = b(h^{-1}w, v)$. With ψ a nontrivial additive character of $\mathbb{F}_q$,

$$N_h = q^{-2} \sum_{\lambda,\mu \in \mathbb{F}_q} \sum_{v \in V} \psi\big(\lambda \mathsf{Q}(v) + \mu F(v)\big).$$

The terms with $\mu = 0$ give $q^{-2} \cdot q \cdot q^6 = q^5$ (Q has q^6 zeros). For $\mu \neq 0$, $\lambda\mathsf{Q} + \mu F = \mu(F - \gamma\mathsf{Q})$ with $\gamma = -\lambda/\mu$, and for fixed μ, γ runs over $\mathbb{F}_q$. The form $R_\gamma = F - \gamma\mathsf{Q}$ has polar form $b(v, (h + h^{-1} - \gamma)w)$, so its rank is $7 - d_\gamma$ with $d_\gamma = \dim\ker(h + h^{-1} - \gamma) = \dim\ker(h^2 - \gamma h + 1)$. For a quadratic form of rank r on $\mathbb{F}_q^7$ (q odd), $|\sum_v \psi(R(v))| = q^{7-r/2}$ (diagonalize; each nonzero coefficient gives a Gauss sum of absolute value $q^{1/2}$). Hence $|N_h - q^5| \le (q-1)\sum_{\gamma \in \mathbb{F}_q} q^{1.5 + d_\gamma/2}$ and $|\mathrm{Perp}(h)| \le \frac{q^5-1}{q-1} + q^{1.5}\sum_\gamma q^{d_\gamma/2}$.

For $\gamma \neq \gamma'$ the polynomials $x^2 - \gamma x + 1$ and $x^2 - \gamma' x + 1$ are coprime (their difference is $(\gamma' - \gamma)x$, and x divides neither), so the kernels are independent and $\sum_\gamma d_\gamma \le 7$. Each $d_\gamma \le 4$:

(i) If $x^2 - \gamma x + 1 = (x - \nu)(x - \nu^{-1})$ with $\nu \in \mathbb{F}_q$, $\nu \neq \pm 1$, the kernel is $\ker(h - \nu) \oplus \ker(h - \nu^{-1})$. An eigenvector v with $hv = \nu v$ has $\mathsf{Q}(v) = \mathsf{Q}(hv) = \nu^2 \mathsf{Q}(v)$, so $\mathsf{Q}(v) = 0$ and $\langle v \rangle$ is a fixed point. So $d_\gamma = 0$.

(ii) $\gamma = \pm 2$: $\ker(h \mp 1)$ contains no nonzero singular vector (it would give a fixed point), so it has dimension ≤ 2 (a quadratic form in ≥ 3 variables over $\mathbb{F}_q$ has a nontrivial zero, by Chevalley–Warning), and $\dim\ker(h \mp 1)^2 \le 2\dim\ker(h \mp 1) \le 4$.

(iii) If $x^2 - \gamma x + 1$ is irreducible, with roots $\nu, \nu^q = \nu^{-1}$ in $\mathbb{F}_{q^2}$, then $d_\gamma = 2\dim_{\mathbb{F}_{q^2}} \ker(h - \nu) \le 2m(\nu)$, with $m(\nu)$ the multiplicity of ν as a root of the characteristic polynomial. By Fact 1.3 the roots are $1, x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}$ with $x_1 x_2 x_3 = 1$. Since $\nu \neq \nu^{-1}$, ν occurs at most once for each index, so $m(\nu) \ge 3$ means $\nu = x_1^{\varepsilon_1} = x_2^{\varepsilon_2} = x_3^{\varepsilon_3}$, hence $\nu^{\varepsilon_1 + \varepsilon_2 + \varepsilon_3} = 1$ with $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 \in \{\pm 1, \pm 3\}$. As $\nu \neq 1$, $\nu^3 = 1$; since $3 \mid q - 1$ the cube roots of unity lie in $\mathbb{F}_q$, contradicting irreducibility. So $m(\nu) \le 2$ and $d_\gamma \le 4$.

The function $d \mapsto q^{d/2}$ is increasing and convex. For integers $0 \le d_\gamma \le 4$ with $\sum d_\gamma \le 7$, raising an entry, or moving one unit from an entry to a larger entry below 4, does not decrease $\sum_\gamma q^{d_\gamma/2}$; so the maximum is at $(4, 3, 0, \ldots, 0)$, where it is $q^2 + q^{1.5} + (q-2)$. This gives the bound. At $q = 7$: $2{,}801 + 907.5 + 343 + 92.6 = 4{,}144.1$. □

Sampled for comparison: $|\mathrm{Perp}(h)| \in \{2{,}752, 2{,}850\}$ for 229 and for 224 fixed-point-free elements of $\mathrm{G}_2(7)$, in two separate samples (`research/omega_repair/logs/fpfperp_q7.txt`, `research/referee/omega-repair/logs/lemma44_sampling_q7.txt`, on their branches).

Theorem 4 (computer-assisted: uses $m_2 \le 140$ from C4)**.** *For $q = 7$, every block pattern, every choice of inner classes and every geometric labelling, ΩH_7 has diameter ≥ 9.*

Proof. By Lemma 5.3(a),(c) we have reading 2 or its dual, in normal form. Take a starved pair (X, c') of the point bar (Lemma 5.6) and suppose that every hub of X is within distance 8 of every point of c'. Consider the 15 elements g_i $(1 \le i \le 5)$ and $g_i^{-1}g_j$ $(i < j)$ of $\mathrm{G}_2(7)$. If one of them fixes a point t, then two of $t, y_1(t), \dots, y_5(t)$ coincide ($g_i(t) = t$, or $g_i(t) = g_j(t)$), and Lemma 5.8 gives a far vertex. So none of them has a fixed point. Then for every point t the six points $t, y_1, \dots, y_5$ are distinct, and by Lemma 5.9 and Corollary 5.5 (at least F far vertices) the configuration at t is of one of the 9 types with $F \le 0$, so at least 6 of the 15 pairs among $t, y_1, \dots, y_5$ are perpendicular. The pair (t, y_i) is perpendicular iff $t \in \mathrm{Perp}(g_i)$, and (y_i, y_j) iff $t \in \mathrm{Perp}(g_i^{-1}g_j)$, since collineations preserve $\perp$. Summing over the 19,608 points t: $\sum_h |\mathrm{Perp}(h)| \ge 6 \cdot 19{,}608 = 117{,}648$ over the 15 elements h, while Lemma 5.10 ($q = 7$, $3 \mid 6$) gives $\sum_h |\mathrm{Perp}(h)| \le 15 \cdot 4{,}144 = 62{,}160$. This is a contradiction. □

Corollary 5.11 (Theorem; computer-assisted at $q = 7, 8, 9$)**.** *For every table value of* ΩH_q — $(10, 8)$ 1,176,480*,* $(11, 8)$ 2,246,940*,* $(12, 8)$ 4,650,100*,* $(14, 8)$ *and* $(15, 8)$ 14,172,480*,* $(16, 8)$ 36,201,060 — *no geometric labelling of the copies gives diameter* 8*, under any reading of Figs. 2.21a/b, any choice of inner classes and any mix of block types. For* $(14, 8)$*,* $(15, 8)$ *and* $(16, 8)$ *the proof is entirely by hand.*

For $q = 7, 8, 11, 13$ (not powers of 3), "any reading" covers reading 1 with both block types only vacuously: by Lemma 5.3(a) no labelling of that reading is geometric, not even the thesis labelling (its transfers map lines to points). The thesis labelling in that reading is covered by Theorem 2.

Computer-checked C5 (consistency)**.** *On explicit graphs in reading 2 with random collineation labellings (a_c, b_c independent), BFS from a hub of a starved class reaches exactly the union of the balls predicted by the route words (referee of the relabelling note):*

q	*graph, source hub*	*far in c' (target type)*	*bound (Cor. 5.5)*	*ecc.*
7	ΩH_7*, aabbb, barred hub (lines)*	*5,221*	–	*9*
8	ΩH_8*, aabbb, barred hub (lines)*	*12,943*	*3,712*	*9*
9	ΩH_9*, aaabbb, unbarred hub (points)*	*22,158*	*3,726*	*9*

With $a_c = b_c$ the distance profile from a hub equals that of the thesis graph (Lemma 5.2).

6. What remains open

(1) **Arbitrary labellings ($q \ge 4$).** Whether some non-geometric labelling of Gómez's wiring gives diameter 8 at $q = 7, 8, 9, 11, 13$ is open; by Corollary 5.11 it would have to be non-geometric. Counting does not exclude it: summed over a hub's routes, the most they can cover is $9.4n$ ($q = 3$), $5.5n$ (4), $5.6n$ (5), $4.3n$ (7), $3.7n$ (9) and $3.1n$ (13), n the number of targets in the copy, so the union bound stays near 3 as q grows (table audit). When a transfer is not geometric, the three terms of a route (walks of length 0, 2, 4 before the pass) can each contribute about q^4 targets instead of collapsing into one ball; a repair would need transfers that scatter the radius-2 and radius-4 balls almost perfectly and fit together across hubs. In every graph where this was checked, all pairs at distance > 8 were pairs (hub, vertex of a copy the hub misses): BFS from all vertices gives exactly twice the count from the hubs (relabelling note, its Computer-checked 2.2, twisted labellings at $q \le 4$; its referee at $q = 3$; the table audit reports the same for every graph it tried). For twisted labellings in reading 2 with both block types present the relabelling note proves this (its Lemma 2.3, with one index corrected by its referee); we do not use or reprove it here.

Computer-checked C6 (searches)**.** *No labelling with diameter* 8 *was found at $q = 4$ (*54,600 *vertices) or $q = 5$ (*195,300*): annealing with a Singer-type symmetry and eight squad variants ended at* 12,075–13,900 *far pairs from* 390 *hub representatives ($q = 4$) and* 35,500–38,000 *from*

252 *($q = 5$) (table audit, round 2). At $q = 7$ a random labelling with the Singer-cube symmetry leaves* 6,813,773 *far pairs from* 2,064 *hub representatives. Mean far-pair counts of uniformly random labellings (table audit):* 1,911 *at $q = 3$ and* 645,753 *at $q = 4$ (all hubs, fully random), and about* 70, $6.55 \cdot 10^6$ *and* $6.5 \cdot 10^8$ *at $q = 2, 5, 7$ (scaled up from symmetric random labellings). Geometric labellings are much worse than random ones at every q tested: annealed collineation labellings leave* 805 *($q = 2$),* 259,169 *(3),* $18.2 \cdot 10^6$ *(4) and* $209 \cdot 10^6$ *(5) far pairs (relabelling note, its Computer-checked 4.7).*

Heuristic H1 (first moment)**.** *If the far pairs of a uniformly random labelling were independent events, the expected number of labellings with diameter* 8 *would be* $\exp(\ln \#\textit{labellings} - E[F])$*, with $(N!)^{10\alpha-2}$ labellings up to renaming the rows of each bar: about $e^{+3.5\cdot10^3}$ at $q = 2$, $e^{+4.8\cdot10^4}$ at $q = 3$, $e^{-4.1\cdot10^5}$ at $q = 4$, $e^{-5.4\cdot10^6}$ at $q = 5$, $e^{-6.4\cdot10^8}$ at $q = 7$. (The table audit used the exponent $10\alpha - 1$; the difference, $\ln N!$, changes no sign.)* Gap: *the far pairs are strongly dependent, and the estimate says nothing about structured labellings.*

(2) **Labellings per (hub class, copy).** If the two classes of one bar meeting a copy may use different bijections, the model of §5.1 does not apply, even when every transfer is a collineation. The counting method fails there for every q: every copy met by X shares a class with c', so there are at least $2\alpha \geq q + 2$ radius-4 centres (at $q = 8$: 10 radius-4 and 318 radius-2 centres, bound −24,448; at $q = 13$: 16 and 864, bound −203,138; computed once, by the referee of the relabelling note, `research/referee/omega-repair/logs/perclass.txt`). Open.

(3) **Other wirings.**

Heuristic H2 (other wirings)**.** *Theorems 3 and 4 use the starved pairs of Gómez's wiring, where only $\alpha \approx q/2$ copies reach c' in one pass. A wiring giving every pair about $q + 3$ one-pass routes is not excluded by size. A near-exact cover would need about q of the centres on one totally singular line through t, for every t (Lemma 5.7: centres in general position overlap in about q^3 points), which points to long root elations $x \mapsto x + b(x, u)v - b(x, v)u$ (u, v spanning a hexagon line); a root group moves t only along one line and has fixed points, which Lemma 5.8 penalises.* Gaps: *"near-exact cover ⇒ line structure" and "line structure for all t ⇒ one root group" are not proved.*

(4) $q \leq 5$ **in the geometric class** (not table values). Neither the count of Theorem 3 nor the $q = 7$ argument works: the general-position count of Lemma 5.9 gives −1,024 at $q = 5$ (and −357 if the same formula is applied at $q = 4$, where Lemma 5.7 is not available because b is degenerate), and Lemma 5.10 needs $3 \mid q - 1$, which fails at $q = 5$. There are only the measurements of C6. The subclass $a_c = b_c$ is settled for every q (Lemma 5.2).

(5) **The relabelled graphs at $q = 2, 3$.**

Computer-checked C7 (diameter 8 at $q = 2, 3$)**.** *With arbitrary per-copy labellings found by annealing (table audit), ΩH_2 (pattern ab,* 1,890 *vertices, copy vertices of degree* 5*, hubs of degree* 4*) and ΩH_3 (pattern aba, reading 1,* 14,560 *vertices,* 6*-regular) have diameter* 8*. For ΩH_3 there are two labellings, equivariant under a Singer-type collineation of order* 13 *(*`omega/labs/q3_aba_LLL_G13.lab` *and* `q3_aba_LLL_G13_Z5psi.lab`*); they were checked by all-pairs BFS with no symmetry assumed and an independent wiring check (audit), and by a third rebuild with separate wiring and BFS code (referee of the first write-up). For ΩH_2 there is one labelling, checked by all-pairs BFS by the audit only.*

These graphs are far below the table at $(5, 8)$ and $(6, 8)$ (9,234 and 29,790); they only show that the bound $D_1 + 2$ can hold for a suitable labelling at $q \leq 3$. Their labellings have not been analysed: we know no description of them, nor whether any structure in them extends to larger q.

7. The table entries: what can be claimed

cell	value	thesis labelling	geometric relabellings	arbitrary
$(10,8)$	1,176,480 $= \lvert\Omega H_7\rvert$	≥ 9; $= 9$ computed	≥ 9 (Thm 4, computer-assisted)	open
$(11,8)$	2,246,940 $= \lvert\Omega H_8\rvert$	≥ 9; ecc. 9 computed	≥ 9 (Thm 3, computer-assisted)	open
$(12,8)$	4,650,100 $= \lvert\Omega H_9\rvert$	≥ 9; $= 9$ computed	≥ 9 (Thm 3, computer-assisted)	open
$(14,8)$	14,172,480 $= \lvert\Omega H_{11}\rvert$	≥ 9; ecc. 9 computed	≥ 9 (Thm 3, by hand)	open
$(15,8)$	14,172,480 $= \lvert\Omega H_{11}\rvert$	≥ 9; ecc. 9 computed	≥ 9 (Thm 3, by hand)	open
$(16,8)$	36,201,060 $= \lvert\Omega H_{13}\rvert$	≥ 9; ecc. 9 computed	≥ 9 (Thm 3, by hand)	open

"Thesis labelling": diameter ≥ 9 by Corollaries 3.1 and 4.1 (every reading); "$= 9$ computed" (for the patterns and pairings of C2(a)) and "ecc. 9 computed" refer to C2. "Geometric relabellings": in reading 1 with both block types there are none unless $q = 9$ (remark after Corollary 5.11).

What can be claimed (the two referees' recommended wording, combined):

> The table entries $(10,8)$ 1,176,480, $(11,8)$ 2,246,940, $(12,8)$ 4,650,100, $(14,8)$ 14,172,480, $(15,8)$ 14,172,480 and $(16,8)$ 36,201,060 are the orders of Gómez's graphs ΩH_q (thesis, UPC 1986, §2.9.3; $q = 7, 8, 9, 11, 11, 13$). The thesis claims diameter $\leq D_1 + 2 = 8$ without proof. As defined there (every copy of the hexagon carries the same labels, and a copy of K_1 with label t is joined only to vertices labelled t), ΩH_q has diameter at least 9 for each of these q, under every reading of Figs. 2.21a/b and every pairing of point labels with line labels (proof: Theorems 1 and 2; also checked by computation). The paper the table cites (Delorme, Gómez, Quisquater, "On large bipartite graphs") was never published, so these cells have no checkable graph of diameter 8.
>
> Give each copy of the hexagon its own labelling: a bijection from the rows to the points and one to the lines, used by every hub class of that colour that meets the copy. Call the labelling geometric if every transfer map between two copies met by one hub class is induced by an automorphism of the incidence graph of $H(q)$ (a collineation, or a duality when $q = 3^e$). For every $q \geq 7$, every geometric labelling gives diameter ≥ 9. This holds in every reading of Figs. 2.21a/b, for every choice of inner classes and every mix of block types. It covers all six table entries. For $q \geq 8$ the proof is a ball-counting argument; it gives explicit numbers: some hub class X and missed copy c' have at least $q^5 - \alpha q^4 - m_2 q^2$ vertices of c' at distance ≥ 9 from every hub of X (3,712, 3,726, 26,862 and 84,162 at $q = 8, 9, 11, 13$). For $q = 7$ the proof adds Bonferroni inequalities and a character-sum bound on how many points a fixed-point-free element of $G_2(7)$ moves to perpendicular points.

Two precisions to this wording. The numbers 3,712, 3,726, 26,862, 84,162 use starved pairs of either hub class (Corollary 5.5 and the line-bar remark after Lemma 5.4). For $q = 7, 8, 11, 13$ the phrase "every reading" covers reading 1 with both block types only vacuously, since no labelling of that reading is geometric there; in that reading the thesis labelling is covered by Theorem 2.

What cannot be claimed.

- That no $(\Delta, 8)$ graph of these orders exists. The cells have "no checkable graph of diameter 8", nothing more.
- That no relabelling of ΩH_q works. Arbitrary per-copy labellings are open for every $q \geq 4$ (and work at $q = 2, 3$); labellings per (hub class, copy) are open; other wirings are open.
- Anything about $q \leq 5$ in the geometric class beyond measurements.
- Anything about $\Omega H_{11}d$ ($N(15,8) \geq 14{,}978{,}080$ in the thesis): it is not defined in §§2.7–2.9, and the table does not use it.
- Anything about the $(13,8)$ entry 5,314,680, also credited to Delorme–Gómez–Quisquater: it is not in the thesis, and nothing here bears on it.

The lab's own graphs in these cells do not depend on any of this: $(11,8)$ 2,258,520 and $(15,8)$ 18,128,880 are explicit Haar graphs (independently checked; see the repository README), and the

largest valid $(12, 8)$ graph we know is a Haar graph on $\mathrm{PSL}(2, 67) \times \mathbb{Z}_{13}$ with 3,909,048 vertices, below the table value (branch `claude/cell-12-8-search-qhi11t`, checked with `build_haar2.py`).

8. Sources, programs and logs

All in the repository github.com/dogaozden/degree-diameter, by branch (commit in brackets).

- *Theorems 1, 2 and the thesis graphs*: `research/squads/cells/d12D8_table.md` and `research/d12D8/omega/` (`hexq6.c`, `omega.c`, `omega_bfs.c`, `apsp_auto.c`, `haut.py`, `omgens.py`, logs), branch `claude/own-12-8-cell-ig5jhd` [568fb4f].
- *Referee of Theorems 1, 2*: `research/referee/omega-dq.md` and `research/referee/omega-dq/` (`hexgen.c`, `hexgf.c`, `bitapsp.c`, `omegaref.c`, `omegacheck.py`, `omegalab.py`, `thm2_bound.py`, logs), branch `claude/omega-dq-referee-bzzlid` [b5db2bb].
- *Relabellings*: `research/theorems/OMEGA_REPAIR.md` and `research/omega_repair/` (`twist.c`, `routecount.c`, `routecount2.c`, `degen.py`, `perpbound.py`, `tanneal.c`, logs; internal $q = 7$ report `referee/REPORT_q7.md`), branch `claude/thm-omega-repair-m4luls` [5f64a1d].
- *Referee of the relabellings*: `research/referee/omega-repair.md` and `research/referee/omega-repair/` (`buildhex.py`, `ghcheck.c`, `collin.py`, `myroutes.py`, `sweep.py`, `perclass.py`, `mytypes.py`, `e2e43.py`, `myperp44.py`, `omg.c`, `driver.py`, `drg.py`, logs), branch `claude/referee-omega-repair-qt17fb` [6337c85].
- *Table audit* (free relabellings at $q = 2, 3$; searches; union bound; first moment): `research/projects/table-audit/AUDIT.md` §§4, 7, `omega/NOTES.md`, `omega/labs/`, `checks/omega_labelled.txt`, `checks/omega_D8.txt`, `checks/witness_omega_thesis.txt`, branch `claude/table-audit-2-p7x6hn` [e1fe451].
- *Lean*: `lean/BipartiteDD/LeanMaxx/Projection/` (`Walk.lean`, `Main.lean`, `Axioms.lean`), `research/lean-maxx/STATUS-projection.md`, branch `claude/leanmaxx-projection-r0kehl` [50cc900]; verification `research/lean-maxx/VERIFY.md`, branch `claude/leanmaxx-verify-9ummw2` [2ff0c8e].
- *This note*: `research/theorems/OMEGA_NOTE.md` and `research/theorems/omega_note/` (this file, `starved.py`, `c4_families.py`, `types43.py`, `typetable.py`, `lem16.py`, logs), branch `claude/omega-d8-note-xrbqux`. Re-run for this note: the constants of Theorem 2 (`thm2_bound.py` of the first referee), the route words for $\alpha = 5, \ldots, 8$ (`starved.py`, with the second referee's `myroutes.py`), Table 1 (`types43.py`, from scratch), and Lemmas 1.5, 1.6 on $H(3)$, $H(4)$ (`lem16.py`); `c4_families.py` was written by an independent reviewer session, which read the draft against all sources; its corrections are applied.

References

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[2] C. Delorme, J. Gómez, J.-J. Quisquater, *On large bipartite graphs*, manuscript (cited as unpublished; not found).

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[4] M. Miller, J. Širáň, Moore graphs and beyond: a survey of the degree/diameter problem, *Electron. J. Combin.* (2013), Dynamic Survey DS14v2.

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